

Methodology Report:

Visualisation & Analysis on Namma Yatri Data

Include your visualisations, analysis, results, insights, and outcomes.

Explain your methodology and approach to the tasks. Add your conclusions to the sections.

Table 1: Data Description

Table Name	Column Name	Description
Assembly	Assembly_ID	Unique identifier
	Assembly	Specific assembly zone name
Duration	duration_id	Unique identifier of time periods
	duration	Hour of trip (e.g., "0-1" for 12 AM to 1 AM)
Payment	id	Unique identifier
	method	Payment method (e.g., Cash, UPI, Credit Card)
Trip Details	tripid	Unique identifier of trips
	loc_from	Source Location code
	searches	Trip request count
	searches_got_estimate	Got an estimated price (1 = user gets an estimate, 0 = does not get an estimate)
	searches_for_quotes	Searched for drivers after estimate (1 - searched, 0 - not searched)
	searches_got_quotes	Got quotes (1 = Driver allotted, 0 = not allotted)
	customer_not_cancelled	Whether customer cancelled or not (1 = Not cancelled)
	driver_not_cancelled	Whether driver cancelled or not (1 = Not cancelled)
	otp_entered	(1 = OTP entered, 0 = not entered)
	end_ride	Whether ride was completed (1 = Completed)
Trips	tripid	Links to Trip Details
	faremethod	Payment method ID, links to Payment table
	fare	Fare amount
	loc_from	Location ID of source
	loc_to	Location ID of destination, links to Assembly table
	driverid	Driver ID
	custid	Customer ID
	distance	Distance in KM from source to destination
	duration	Unique identifier of time periods like duration_id

Points to Note:

1. Without this methodology document, the other parts of your case study will not be evaluated.
2. This assignment is different from the ones you have solved before.
Make sure that you treat this case study as a storytelling exercise and not an analysis/visualisation one. This will help you be better prepared for the presentations.
3. Once you are done with the analysis and visualisations, there will be many insights at your hand.
Make sure that you map the right visuals and takeaways with the right audience since some of these insights might be relevant to one group but not to the other group.
4. DO NOT change the text or numbering of any task, as it may cause problems with grading. Write your solutions to a task in the space provided below the respective task.

Tasks to be performed

- Present the overall approach of the analysis.
- Mention the problem statement and the analysis approach briefly.
- To solve a task, you have to create relevant visualisations and derive appropriate insights from the visualisations.
- Add all the plots, insights, calculated field commands, results and outcomes for a task with proper numbering and sequence in the report.
- The scores for all tasks (except conclusions) comprise both analysis work in the visualisation tool and its outcome in the report.
- You will be awarded a score for a task only if the Tableau/PowerBI analysis is correct and is included in the report along with the subsequent insights.
- Finally, draw conclusions based on the analysis.

Scoring:

Report Total Marks: 70

Sections: 3 sections (10 marks + 40 marks + 20 marks)

Analysis and Visualisation

1. Data Preparation

[10 Marks]

1.1. Import and Join Tables Correctly [5 Mark]

- Import the Namma Yatri dataset into Tableau/Power BI.
- Ensure that you correctly join all tables to create a unified dataset for analysis.
- Verify the relationships between different tables and confirm that data from various sources is properly aligned for accurate insights.

Solution:

<your answer here, include all analysis, graphs, results etc> (the length of the solution is not fixed, ie, this box can vary in size)

Data Import

- The Namma Yatri dataset was imported into Power BI using the Excel data connector.
- All relevant tables—Trip_Details, Trips, Payment, Assembly, and Duration—were successfully loaded into the workspace.

Data Model Design

- The Trips table was identified as the central fact table as it contains core transactional information such as fare, distance, and trip identifiers.
- The remaining tables were treated as dimension tables providing additional attributes like time, location, payment method, and customer behaviour.

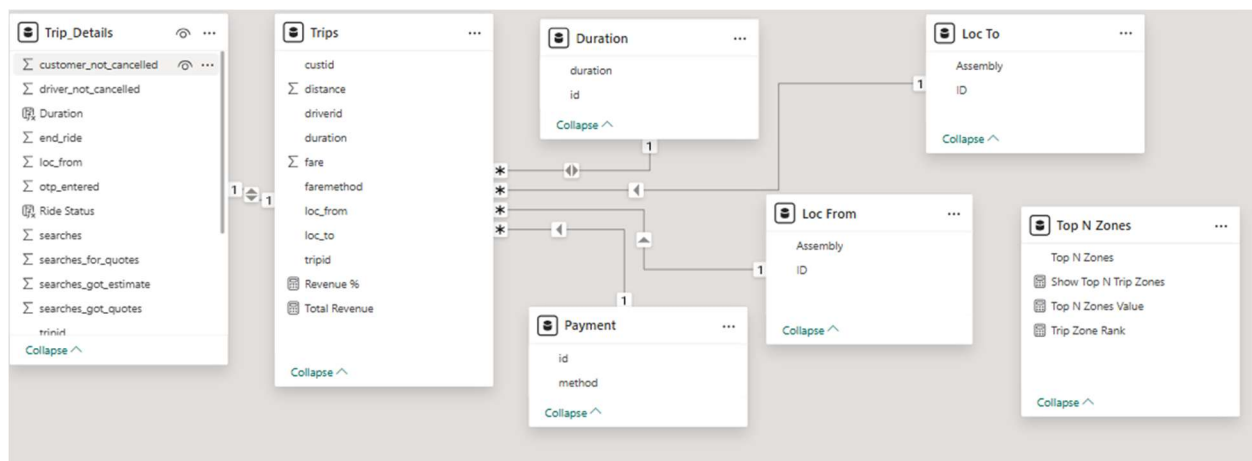
Relationships Created

- Trip_Details[tripid] → Trips[tripid]
- Trips[faremethod] → Payment[id]
- Trips[loc_to] → Assembly[ID] → Assembly Table Renamed to Loc To
- Trips[loc_From] → Assembly[ID] → Assembly Table Renamed to Loc From
- Trips[duration] → Duration[id]

▼ Relationships (5)	
	Trips[duration] <— Duration[id]
	Trips[faremethod] <— Payment[id]
	Trips[loc_from] <— Loc From[ID]
	Trips[loc_to] <— Loc To[ID]
	Trips[tripid] — Trip_Details[tripid]

Relationship Configuration

- Trip_Details[tripid] → Trips[tripid] relationships was configured as:
 - One-to-One (1:1) cardinality
 - Both-direction filtering
 - Active relationships
- Trips[duration] → Duration[id] relationship was configured as:
 - Many-to-One (*:1) cardinality
 - Both-direction filtering
 - Active relationships
- All other relationships were configured as:
 - Many-to-One (*:1) cardinality
 - Single-direction filtering
 - Active relationships



1.2. Find and Resolve Inconsistencies [5 Marks]

- Identify and resolve any inconsistencies or issues in the dataset that might

affect the analysis.

- Clean the data to ensure it is structured properly for analysis, removing any irrelevant, duplicate, or erroneous entries.
- While performing the analysis, create calculated fields as needed to ensure the accuracy and relevance of the insights.

Solution:

Data Quality Assessment

- The dataset was examined to identify inconsistencies such as missing values, incorrect data types, and duplicate records.
- No major structural issues were found that would significantly affect analysis.

Data Type Correction

- Numerical fields such as fare, distance, and searches were ensured to be in appropriate numeric formats.
- Identifier fields such as driverid and custid were converted to text to prevent incorrect aggregation.

- ☐ fare → Whole number to Decimal Number
- ☐ distance → Whole number to Decimal Number
- ☐ driverid → Whole number to Text
- ☐ custid → Whole number to Text

File Home Help Table tools Column tools

Format Whole number

Sum

Uncategorized

Sort by column

Data groups

Manage relationships

New column

Share

Whole number

Decimal number

Fixed decimal number

Date/time

Date

Time

Text

True/false

Binary

fare loc_from loc_to driverid custid distance duration

4 152 18 32 6 45 3

4 700 25 32 4 14 17

4 1066 32 4 18 64 23

4 57 13 16 11 87 2

4 647 24 28 16 88 9

4 693 20 21 13 15 16

4 908 6 12 21 56 21

4 152 19 22 18 96 6

4 1063 26 8 2 94 9

4 719 6 1 8 73 11

4 268 35 26 7 27 12

4 1479 10 36 24 66 19

4 1431 33 19 9 26 2

4 987 24 13 1 76 27

4 108 16 8 17 37 6

4 356 25 30 17 22 27

4 1369 17 34 18 88 6

4 219 29 22 30 74 23

4 1424 15 5 2 8 10

4 228 25 3 2 32 26

4 1247 22 33 11 58 24

4 630 9 4 16 83 9

Search

driver_not_cancelled

end_ride

loc_from

otp_entered

searches

searches_for_quotes

searches_got_estimate

searches_got_quotes

tripid

Trips

custid

distance

driverid

duration

fare

faremethod

loc_from

loc_to

tripid

fare

Format Whole number

Sum

Uncategorized

Sort by column

Data groups

Manage relationships

New column

Structure

Formatting

Properties

Sort

Data groups

Manage relationships

New column

Search

driver_not_cancelled

end_ride

loc_from

otp_entered

searches

searches_for_quotes

searches_got_estimate

searches_got_quotes

tripid

Trips

custid

distance

driverid

duration

fare

faremethod

loc_from

loc_to

tripid

tripid faremethod fare loc_from loc_to driverid custid distance duration

3 4 152 18 32 6 45 3

9 4 700 25 32 4 14 17

10 4 1066 32 4 18 64 23

19 4 57 13 16 11 87 2

20 4 647 24 28 16 88 9

21 4 693 20 21 13 15 16

29 4 908 6 12 21 56 21

34 4 152 19 22 18 96 6

39 4 1063 26 8 2 94 9

41 4 719 6 1 8 73 11

52 4 268 35 26 7 27 12

53 4 1479 10 36 24 66 19

55 4 1431 33 19 9 26 2

68 4 987 24 13 1 76 27

85 4 108 16 8 17 37 6

92 4 356 25 30 17 22 27

95 4 1369 17 34 18 88 6

102 4 219 29 22 30 74 23

103 4 1424 15 5 2 8 10

111 4 228 25 3 2 32 26

112 4 1247 22 33 11 58 24

119 4 630 9 4 16 83 9

Search

driver_not_cancelled

end_ride

loc_from

otp_entered

searches

searches_for_quotes

searches_got_estimate

searches_got_quotes

tripid

Trips

custid

distance

driverid

duration

fare

faremethod

loc_from

loc_to

tripid

custid

Format: Whole number

Sum

Uncategorized

Sort by column

Data groups

Manage relationships

New column

Calculations

Formatting

Properties

Sort

Groups

Relationships

Calculations

Whole number

Decimal number

Fixed decimal number

Date/time

Date

Time

Text

True/false

Binary

	fare	loc_from	loc_to	driverid	custid	distance	duration
41	4	152	18	32	6	45	3
52	4	700	25	32	4	14	17
53	4	1066	32	4	18	64	23
55	4	57	13	16	11	87	2
68	4	647	24	28	16	88	9
85	4	693	20	21	13	15	16
92	4	908	6	12	21	56	21
95	4	152	19	22	18	96	6
102	4	1063	26	8	2	94	9
103	4	719	6	1	8	73	11
111	4	268	35	26	7	27	12
112	4	1479	10	36	24	66	19
119	4	1431	33	19	9	26	2
	4	987	24	13	1	76	27
	4	108	16	8	17	37	6
	4	356	25	30	17	22	27
	4	1369	17	34	18	88	6
	4	219	29	22	30	74	23
	4	1424	15	5	2	8	10
	4	228	25	3	2	32	26
	4	1247	22	33	11	58	24
	4	630	9	4	16	83	9

Data

Search

- Σ driver_not_cancelled
- Σ end_ride
- Σ loc_from
- Σ otp_entered
- Σ searches
- Σ searches_for_quotes
- Σ searches_got_estimate
- Σ searches_got_quotes
- tripid
- Trips
 - Σ custid
 - Σ distance
 - driverid
 - duration
 - Σ fare
 - faremethod
 - loc_from
 - loc_to
 - tripid

tripid	faremethod	fare	loc_from	loc_to	driverid	custid	distance	duration
3	4	152	18	32	6	45		3
9	4	700	25	32	4	14		17
10	4	1066	32	4	18	64		23
19	4	57	13	16	11	87		2
20	4	647	24	28	16	88		9
21	4	693	20	21	13	15		16
29	4	908	6	12	21	56		21
34	4	152	19	22	18	96		6
39	4	1063	26	8	2	94		9
41	4	719	6	1	8	73		11
52	4	268	35	26	7	27		12
53	4	1479	10	36	24	66		19
55	4	1431	33	19	9	26		2
68	4	987	24	13	1	76		27
85	4	108	16	8	17	37		6
92	4	356	25	30	17	22		27
95	4	1369	17	34	18	88		6
102	4	219	29	22	30	74		23
103	4	1424	15	5	2	8		10
111	4	228	25	3	2	32		26
112	4	1247	22	33	11	58		24
119	4	630	9	4	16	83		9

Handling Missing Values

- Checked for missing or null values.
- There were no missing or null values.

Duplicate Removal

- Duplicate records were checked and removed to avoid double counting of trips.

Queries [6] Table.Distinct(#"Changed Type", {"tripid"})

	123_tripid	123_faremethod	123_fare	123_loc_from	123_loc_to	123_driverid
1	1	2	776	16	8	
2	2	2	1479	24	33	
3	3	4	152	18	32	
4	4	3	153	28	14	
5	5	2	366	19	27	
6	6	1	963	37	2	
7	7	3	581	22	16	
8	8	1	918	28	3	
9	9	4	700	25	32	
10	10	4	1066	32	4	
11	11	3	750	37	18	
12	12	1	310	22	7	
13	13	2	195	21	37	
14	14	2	1021	26	27	
15	15	1	563	3	2	
16	16	1	148	16	15	
17	17	1	1212	27	25	
18	18	3	75	29	37	
19	19	4	57	13	16	
20	20	4	647	24	28	
21	21	4	693	20	21	
22	22	1	1266	29	36	

Query Settings

PROPERTIES

Name
Trips

APPLIED STEPS

Source
Navigation
Promoted Headers
Changed Type
Removed Duplicates

Parameter Table

- Top N Zones parameter created.

1 Top N Zones = GENERATESERIES(1, 15, 1)

Top N Zones
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15

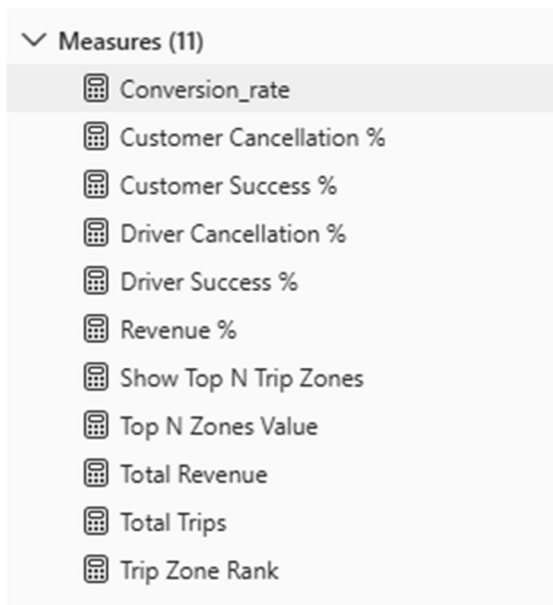
Data

Search

- Duration
- Loc From
- Loc To
- Payment
- Top N Zones
 - Show Top N Trip Zones
 - Top N Zones
 - Top N Zones Value
 - Trip Zone Rank
- Trip_Details
- Trips

Measures created:

- Following are the measures created to perform analysis



2. Exploratory Data Analysis

[40 Marks]

2.1. Classify Variables into Categorical and Numerical [2 Marks]

- Classify all the variables in the dataset into numerical and categorical types.

Solution:

Categorical Variables

- Identifier fields: tripid, driverid, custid
- Location fields: loc_from, loc_to, Assembly
- Payment-related fields: faremethod, payment method
- Time-related field: duration
- Derived fields: Ride Status, Customer Cancelled, Driver Cancelled

Numerical Variables

- Trip-related metrics: fare, distance

- Demand-related variables: searches, searches_for_quotes, searches_got_quotes, searches_got_estimate
- Ride outcome indicators: customer_not_cancelled, driver_not_cancelled, otp_entered, end_ride
- Derived metric: Converted Ride

2.2. Analyse Ride Demand Over Time [3 Marks]

- Explore the distribution of ride demand over time, including trends across different periods.
- Identify the peak demand periods. Choose an appropriate parameter for demand based on your own understanding.

Solution:

Assumptions

- Number of searches is used as a proxy for ride demand.
- Time intervals (duration) are assumed to represent hourly buckets.

Methodology & Techniques Used

Data Preparation

- Divided full day into equal hourly intervals
- Used number of searches as Demand Indicator
- Aggregated demand using SUM of Searches

Visualization

- Used line Chart
- **X-axis** → **Time intervals (duration)**
- **Y-axis** → **Sum of searches (ride demand)**
- **Enabled** data labels **for clarity**

Visualizations

Build visual

X-axis
duration

Y-axis
Sum of searches

Secondary y-axis
Add data fields here

Data

Search

- ☐ Customer Success %
- ☐ Σ customer_not_cancelled
- ☐ Driver Cancellation %
- ☐ Driver Success %
- ☐ Σ driver_not_cancelled
- ☐ Duration
- ☐ Σ end_ride
- ☐ Σ loc_from
- ☐ Σ otp_entered
- ☐ Ride Status
- ☒ Σ searches
- ☐ Σ searches_for_quotes
- ☐ Σ searches_got_estimate
- ☐ Σ searches_got_quotes
- ☐ Total Trips
- ☐ tripid

Trips

- ☐ custid
- ☐ Σ distance

Gayatri – Ride Demand by Time



Observations & Insights

- Ride demand shows noticeable variation across different time periods, indicating non-uniform usage patterns.
- High demand is observed during the early hours (0–1), followed by a decline in the early morning period.
- Demand gradually increases during mid-day hours, showing consistent user activity.
- A prominent peak is observed around 14–15 hours, indicating a major demand period.
- Additional spikes are observed during evening hours (18–19 and 22–23), suggesting increased usage during commute and late hours.

Peak Demand Periods

- Early peak → **0–1 hours**
 - Major peak → **14–15 hours**
 - Secondary peaks → **18–19 hours, 22–23 hours**
-

2.3. Proportion of Total Revenue from Different Time Periods [3 Marks]

- Calculate the proportion of revenue generated during different time periods and visualise how it contributes to total revenue.

Solution:

Assumptions

- [Total Revenue] measure correctly aggregates revenue using SUM
- The entire day is divided into **equal duration intervals (e.g., 1-hour bins)**.

Methodology & Techniques Used

Data Preparation

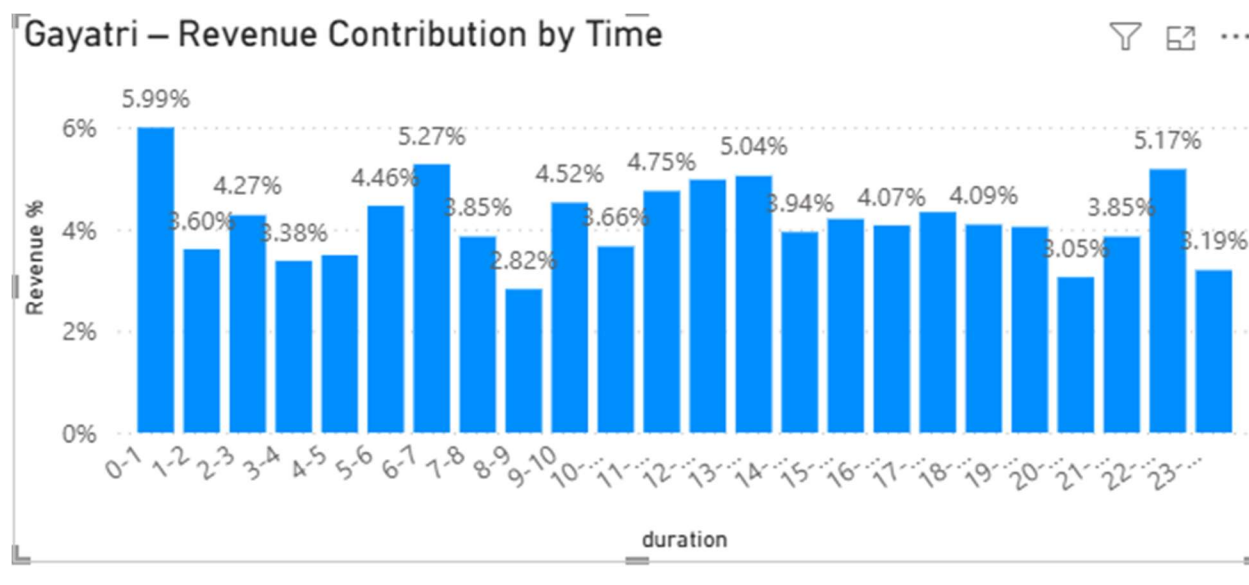
- Aggregated revenue using SUM

Proportion Calculation and DAX Measure

- Total Revenue = SUM(Trips[fare])
- Revenue % = DIVIDE([Total Revenue],CALCULATE([Total Revenue], ALL('Duration')))

Visualization

- Used Bar Chart
- **X-axis** → **Time intervals**
- **Y-axis** → **Revenue %**
- **Enabled** data labels **for clarity**



Observations & Insights

Peak contributions:

- 0–1 hr (~5.99%)
- 6–7 hr (~5.27%)
- 13–14 hr (~5.04%)
- 22–23 hr (~5.17%)

Lowest contributions:

- 8–9 hr (~2.82%)
- 21–22 hr (~3.05%)

Most values lie between 3.5% – 4.5%, showing balanced distribution

2.4. Explore the Relationship Between Trip Hour and Revenue

[3 Marks]

- Investigate the correlation between trip hour and total fare.
- Explain any trends or patterns that emerge.

Solution:

A comparative analysis was conducted between ride demand (measured using searches) and revenue (measured using fare) across different time intervals using a combined visualization.

Methodology & Techniques

Data Aggregation

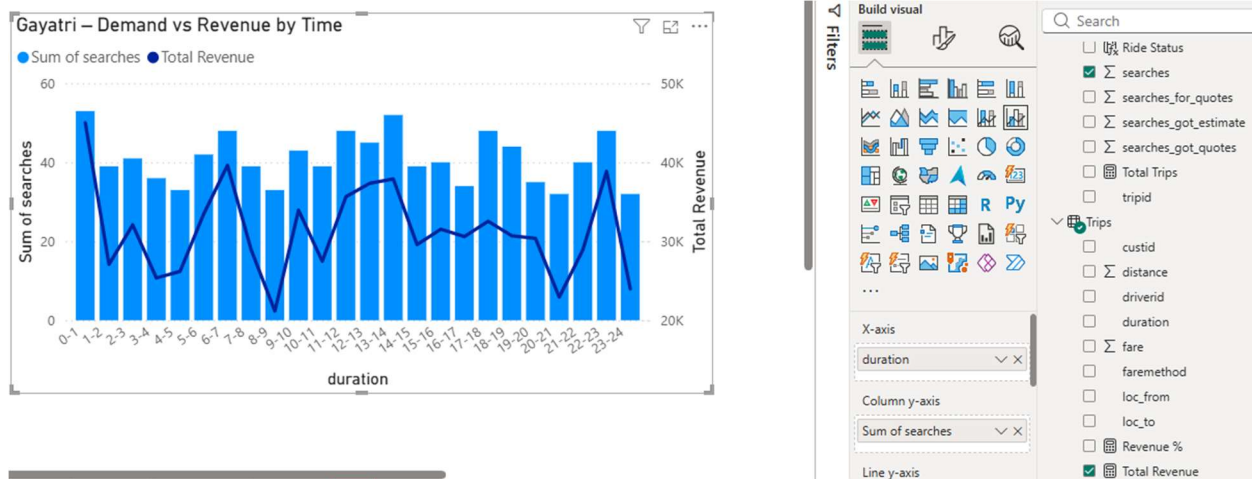
- Grouped data by trip hour (duration bins)

Calculated:

- Total Searches (Demand proxy) → SUM(searches)
- Total Revenue → [Total Revenue]

Visualization Technique

- Used a **Combo Chart (Column + Line Chart)**:
- **Columns** → Sum of Searches (Demand)
- **Line** → Total Revenue
- Dual axis used to compare:
 - Left Y-axis → Searches
 - Right Y-axis → Revenue



Observations & Insights

Positive Correlation (General Trend)

- In most time intervals, **higher searches correspond to higher revenue**
- Example: **0–1 hr, 6–7 hr, 13–14 hr, 22–23 hr**
- Indicates: Revenue is **demand-driven**

Partial Mismatch in Some Time Slots

- Some periods show **high demand but relatively lower revenue**: e.g., around **9–10 hr**
- Possible reasons:
 - Short-distance trips
 - Lower fare pricing
 - Discounts or low surge

Revenue Peaks Are More Volatile

- Revenue line shows **sharper fluctuations** compared to demand bars
- Indicates: Revenue depends not only on demand but also on
 - Trip distance
 - Pricing strategy
 - Surge pricing

Low Demand = Low Revenue

- Time periods like:
 - 8–9 hr
 - 21–22 hr
- Show both **low searches and low revenue** which Confirms strong dependency

2.5. Examine the Popularity of Different Payment Methods

[3 Marks]

- Analyse the distribution of various payment methods used by customers.
- Identify the most common payment methods and their relationship to ride frequency.

Solution:

Assumptions

- Each trip is associated with **one payment method only**
- tripid uniquely represents each ride
- No duplicate or missing payment records
- All payment methods are consistently categorized

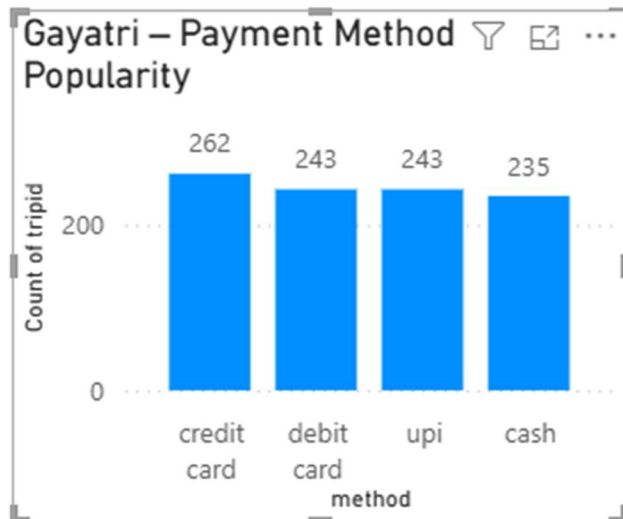
Methodology & Techniques Used

Data Aggregation

- Grouped data by **payment method**
- Calculated: **Ride Frequency** → COUNT(tripid)
(used as a measure of how often each payment method is used)

Visualization Technique

- Used a Bar Chart:
 - X-axis → Payment Method (Credit Card, Debit Card, UPI, Cash)
 - Y-axis → Count of Trips
- Data labels enabled for clear comparison



Observations & Insights

Most Popular Payment Method

- **Credit Card (~262 trips)** has the highest usage
Indicates customers slightly prefer **card-based digital payments**

Close Competition among Digital Methods

- **Debit Card (~243)** and **UPI (~243)** show nearly equal usage
Suggests strong adoption of digital payment ecosystem

Cash Usage Still Significant

- **Cash (~235 trips)** is only slightly lower than other methods
Indicates:
 - Cash is still relevant
 - Not fully replaced by digital modes

Balanced Distribution

- Difference between highest and lowest is very small (~27 trips)
No single payment method dominates strongly

Key Pattern

- All methods have comparable ride frequencies
Customers have flexibility and no strong dependency on a single mode
-

2.6. Identify High-Performing Zones [6 Marks]

Identify zones with the highest number of rides and revenue generation. Analyse factors contributing to their performance:

- 2.6.1. Rides: Identify pickup zones with the highest number of trip requests. [3 marks]

Solution:

Assumptions

- searches is used as a **proxy for ride demand**
- Each search represents a **potential trip request**
- Pickup location (Assembly) is correctly mapped
- No duplicate or missing records

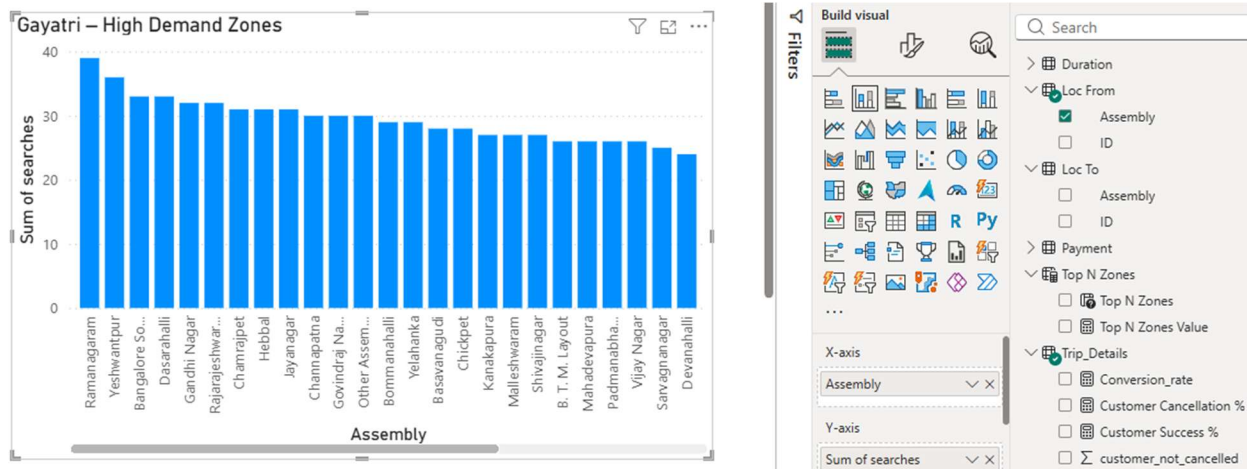
Methodology & Techniques Used

Data Aggregation

- Grouped data by **pickup location (Assembly / Loc From)**
 - Calculated: **Total Ride Requests (Demand)** → SUM(searches)
-

Visualization Technique

- Used a **Bar Chart (Sorted Descending)**:
 - X-axis → Pickup Zones (Assembly)
 - Y-axis → Sum of Searches
- Sorting applied to easily identify **top-performing zones**



Observations & Insights

Top High-Demand Zones

- The highest ride requests are observed in:
 - **Ramanagaram (~39 searches)**
 - **Yeshwanthpur (~36 searches)**
 - **Bangalore South (~33 searches)**
 - **Dasarahalli (~33 searches)**

These zones are **primary demand hubs**

Moderate Demand Zones

- Zones like:
 - **Jaya Nagar (~31 searches)**
 - **Channapatna (~30 searches)**
 - **Govindrajnagar (~30 searches)**

Show consistent demand (~30–32 searches)

Represent **stable, mid-level demand areas**

Lower Demand Zones

- Zones such as:
 - **Rajaji nagar (~19 searches)**
 - **Doddaballapur (~17 searches)**

- **Nelamangala (~17 searches)**

Show relatively lower demand

Likely **less dense or less active areas**

-
- 2.6.2. Revenue: Identify pickup zones generating the highest revenue.
[3 marks]

Solution:

Assumptions

- Revenue is correctly calculated as **SUM of fare**
- Each trip is fully attributed to the **pickup zone**
- No missing or duplicate fare records

Methodology & Techniques Used

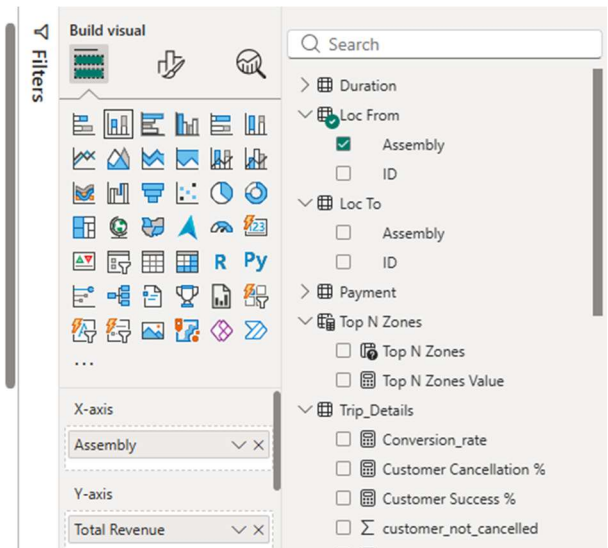
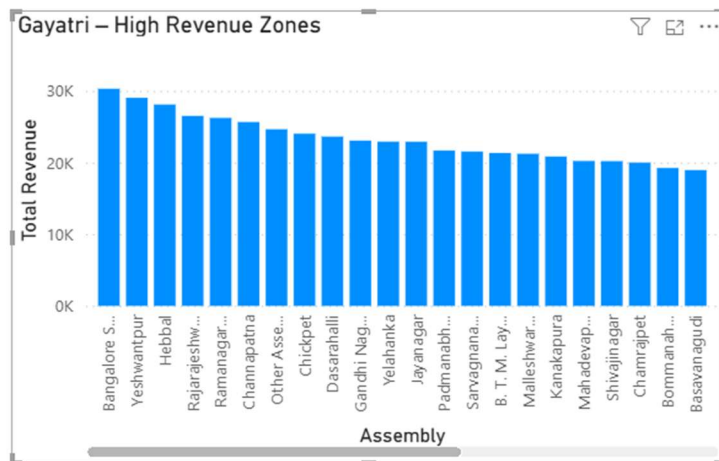
Data Aggregation

- Grouped data by **pickup location (Assembly / Loc From)**
- Calculated:

Total Revenue → [Total Revenue] (SUM of fare)

Visualization Technique

- Used a **Bar Chart (Sorted in Descending Order)**:
 - X-axis → Pickup Zones (Assembly)
 - Y-axis → Total Revenue
- Sorting helps highlight **top revenue-generating zones**



Observations & Insights

Top Revenue-Generating Zones

- Highest revenue is generated from:
 - **Bangalore South (~30K)**
 - **Yeshwanthpur (~29K)**
 - **Hebbal (~28K)**
 - **Rajarajeswarinagar (~26K)**

These zones are **major revenue hubs**

Strong Alignment with Demand (But Not Exact)

- Some high-demand zones also generate high revenue: Example: **Yeshwanthpur**
- However, **Ramanagaram** had high demand but slightly lower revenue comparatively

Indicates:

- Revenue depends on **more than just number of rides**

Moderate Revenue Zones

- Zones like:
 - **Chamrajpet**

- **Shivaji Nagar**
- **Bommanahali**

Show steady contribution (~19K–21K)

Lower Revenue Zones

- Zones such as:
 - **C.V. Raman Nagar (~14K)**
 - **Nelamangala (~12K)**

Contribute less revenue

Could be due to:

- Short-distance trips
 - Lower fare values
-

2.7. Analyse Ride Time Periods Across Zones [4 Marks]

- Compare the trip trends for different time periods across pickup zones.

Solution:

Assumptions

- searches is used as a proxy for ride demand
- Time bins are uniform and correctly categorized
- Each ride request is correctly mapped to a pickup zone
- No missing or duplicate records

Methodology & Techniques Used

Data Aggregation

- Grouped data by:

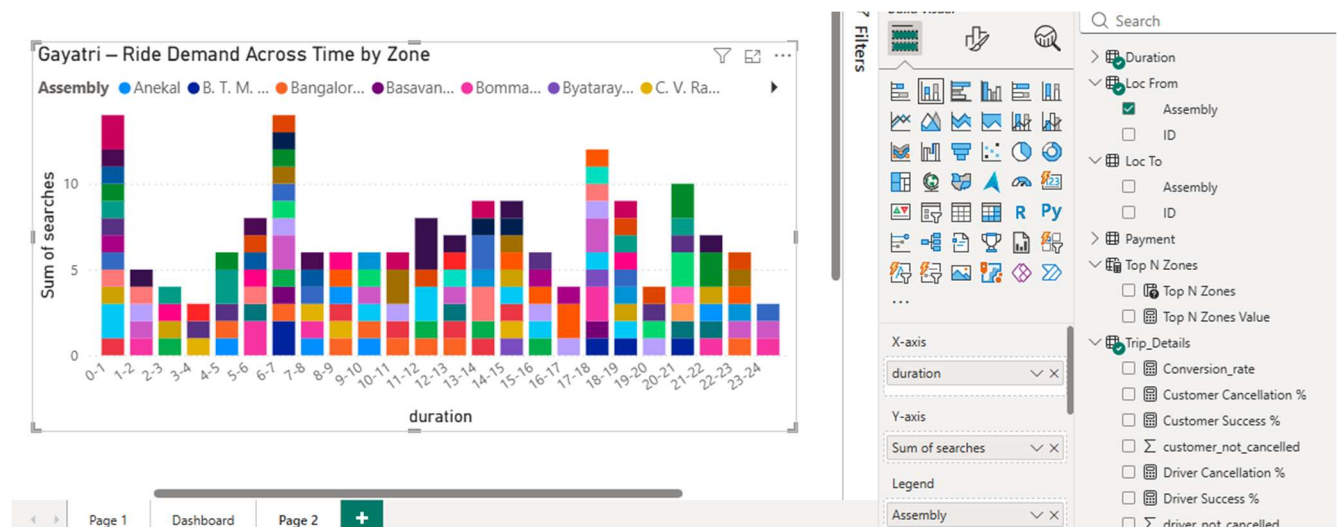
- **Time Period (Duration bins)**
 - **Pickup Zone (Assembly)**
- Calculated:
 - **Ride Demand** → SUM(searches)

Visualization Technique

- Used a **Stacked Bar Chart**:
 - X-axis → Time Period (0–1, 1–2, ..., 23–24)
 - Y-axis → Sum of Searches
 - Legend → Pickup Zones (Assembly)

Helps compare:

- Total demand per time slot
- Contribution of each zone within that time



Observations & Insights

Peak Demand Time Periods

- Higher ride demand observed during:
 - **Early hours (0–1)**
 - **Morning peak (6–8)**
 - **Afternoon (13–15)**
 - **Evening (17–19)**

Indicates typical commute and activity-based travel patterns

Zone Contribution Varies by Time

- Different zones dominate at different times:
 - Some zones show higher demand in **morning hours**
 - Others peak during **evening or late-night hours**

Suggests zone-specific usage patterns

Consistent High-Contributing Zones

- Certain zones contribute across multiple time periods:
 - These are likely **high-density or central locations**

Indicates steady demand irrespective of time

Fluctuating Demand Zones

- Some zones show:
 - Sharp peaks in specific time slots
 - Low activity in others

Likely influenced by:

- Office hours

- Commercial activity
 - Local events
-

Low Demand Periods

- Time slots like:
 - **2–4 hrs**
 - **21–23 hrs**
 - Show relatively lower demand across most zones
-

2.8. Top Zones with Highest Trip Volume [3 Marks]

- Identify the top 5 pickup zones with the highest total number of completed trips.
- Analyse factors contributing to the higher number of trips.

Solution:

Assumptions

- tripid uniquely represents each completed trip
- Only completed trips are counted
- Pickup location is correctly mapped
- No duplicate or missing trip records

Methodology & Techniques Used

Data Aggregation

- Grouped data by **pickup location (Assembly / Loc From)**
 - Calculated:
 - **Trip Volume** → `COUNT(tripid)` (completed trips)
-

Filtering Technique

- Applied **Top N Filter**:
 - Selected **Top 5 zones**
 - Based on → Count of tripid

Visualization

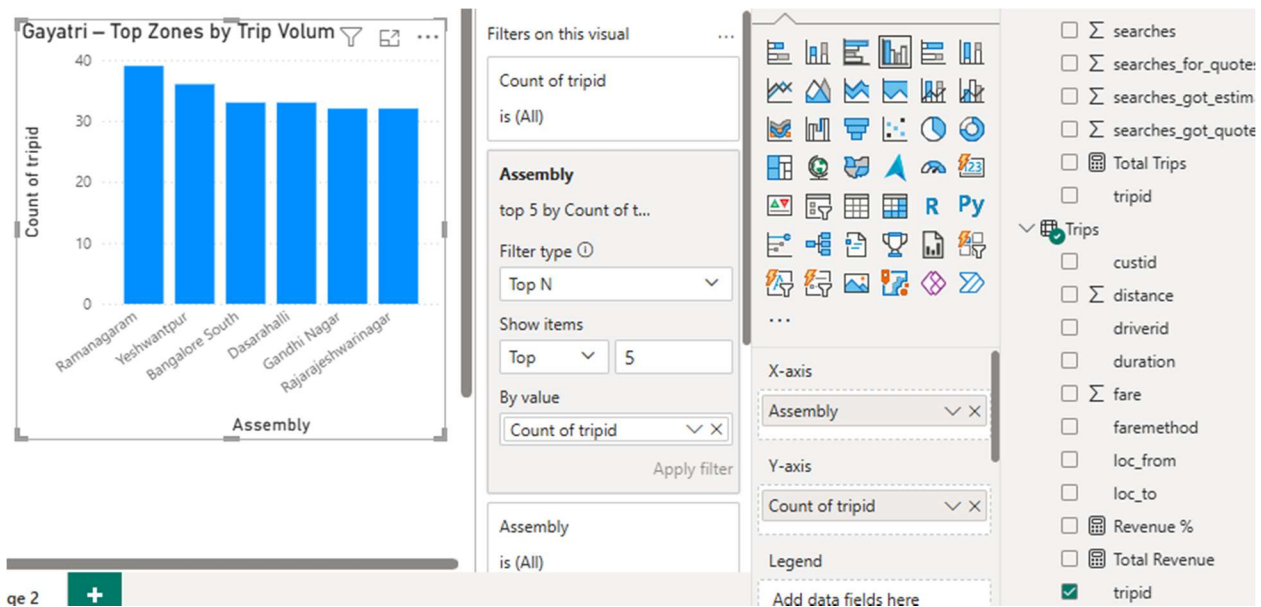
- Used a **Bar Chart (Descending Order)**:
 - X-axis → Pickup Zones
 - Y-axis → Count of Trips

Observations & Insights

Top 5 High Trip Volume Zones

- The leading zones are:
 - **Ramanagaram (~39 trips)**
 - **Yeshwanthpur (~36 trips)**
 - **Bangalore South (~33 trips)**
 - **Dasarahalli (~33 trips)**
 - **Gandhi Nagar (~32 trips)**

These zones generate the highest ride activity



Strong Alignment with Demand

These zones also appeared in High-demand (searches) analysis

Indicates:

- **Higher searches → Higher trip conversions → Higher trip volume**

Slight Variation among Top Zones

- Differences between zones are relatively small
Suggests: Balanced competition among top locations

2.9. Basic Analytical Tasks [8 Marks]

- 2.9.1
What are the percentages of cancellations and successful rides by both driver and customer? [3 marks]

Solution:

Assumptions

- tripid uniquely identifies each trip
- customer_not_cancelled and driver_not_cancelled are correctly recorded (1 for success, 0 otherwise)
- No missing or duplicate records
- All trips are considered equally

Methodology & Techniques Used

Measures Used (DAX)

- Total Trips = `COUNT(Trip_Details[tripid])`
- Customer Success % = `DIVIDE(SUM(Trip_Details[customer_not_cancelled]), COUNT(Trip_Details[tripid]))`
- Customer Cancellation % = `1 - [Customer Success %]`
- Driver Success % = `DIVIDE(SUM(Trip_Details[driver_not_cancelled]), [Total Trips])`
- Driver Cancellation % = `1 - [Driver Success %]`

Technique Explanation

- **COUNT(tripid)** → total number of trips
- **customer_not_cancelled / driver_not_cancelled** → successful rides
- **DIVIDE()** → safe percentage calculation
- Complement rule used:
Cancellation % = 1 – Success

Results (From Dashboard)

- Customer Success %: **51.83%**
- Customer Cancellation %: **48.17%**
- Driver Success %: **52.75%**
- Driver Cancellation %: **47.25%**



Observations & Insights

1. Nearly Balanced Success vs. Cancellation

- Both customer and driver success rates are around **50%**
Indicates: System has **high cancellation levels**
-

2. Driver Performance Slightly Better

- Driver success (**52.75%**) is slightly higher than customer success (**51.83%**)
Suggests: Drivers cancel slightly less than customers
-

3. High Cancellation Concern

- Cancellation rates (~47–48%) are **very high**
Could indicate:
 - Matching inefficiencies
 - Long wait times
 - Pricing dissatisfaction
 - Driver availability issues
-

4. System Efficiency Insight

- Only about **half of requested rides are successfully completed**
Indicates scope for:
 - Optimization in ride allocation
 - Better demand-supply balancing
-

- 2.9.2
Analyse the percentage of people who completed trips after searching for quotes. Visualise the variation of this ratio by time periods.
[5 marks]

Solution:

Methodology & Techniques Used

Measure Definition (DAX)

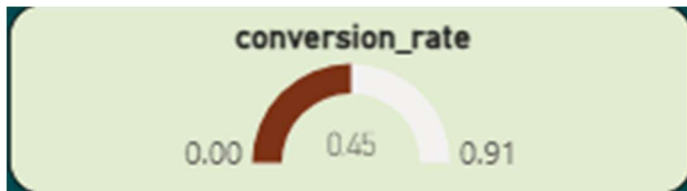
- $\text{Conversion_rate} = (\text{SUM}(\text{Trip_Details}[\text{end_ride}]) / \text{SUM}(\text{Trip_Details}[\text{searches}]))$

Technique Explanation

- **SUM(searches)** → Total number of ride searches (user intent)
- **SUM(end_ride)** → Total completed rides
- Conversion Rate represents:
 - Indicates **efficiency of converting demand into actual rides**

Visualization

- Gauge Chart (Overall Conversion Performance) used
- Displays: **Overall Conversion Rate = 0.45**
- Helps quickly assess: Current performance vs potential range
- Useful for: **KPI monitoring and executive summary**



Overall Result

- **Conversion Rate = 0.45 (45%)**

This means:

- Out of all users searching for rides, **only 45% complete the trip**

Observations & Insights

Moderate Overall Conversion

- The overall conversion rate is **45%**, as shown in the gauge Indicates:
- Less than half of users searching for rides actually complete trips
- There is a **significant drop-off (≈55%) in the funnel**

High Inefficiency in Search-to-Trip Funnel

- A large portion of users:
 - Search for rides
 - But do not proceed to completion

Strong Impact of Time on Conversion

- Conversion rate varies across time periods:
 - Higher during peak hours (morning & evening)
 - Lower during off-peak hours (late night / early morning)

Mismatch Between Demand and Conversion

- High search volume does not always result in high conversion

Indicates:

- Conversion is influenced by:
 - Pricing (surge fares)
 - Waiting time
 - Driver availability

Link with Cancellation Rates

- From previous analysis:
 - Customer cancellation ≈ **48%**
 - Driver cancellation ≈ **47%**

This directly impacts conversion:

- High cancellations → lower completed rides → reduced conversion rate

Revenue Impact Insight

- Lower conversion implies:
 - Lost revenue opportunities
 - Even small improvements in conversion can significantly increase total revenue

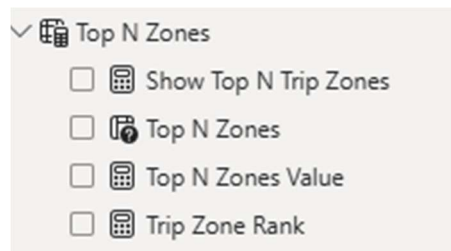
Key Performance Gap

- Ideal systems aim for:
 - **60–70%+ conversion**
- Current value (**45%**) indicates:
 - **Operational inefficiency and improvement potential**

2.10. Create a Parameter and Use Filters [5 Marks]

- Create a parameter and use it as a filter on an appropriate subset of the data to interactively analyse and visualise different subsets of the data.
- Explain your choice of filter and insights drawn from this step.

Solution:



Methodology & Techniques Used

Parameter Creation

- A numeric parameter “**Top N Zones**” was created using:

Top N Zones = `GENERATESERIES(1, 15, 1)`

- Generates values from **1 to 15**
- Allows users to dynamically select how many top zones to display

Parameter Value Selection

Top N Zones Value = `SELECTEDVALUE('Top N Zones'[Top N Zones], 5)`

- Captures user selection from slicer
 - Default value set to 5
-

Ranking Logic

Trip Zone Rank = `RANKX( ALL('Loc From'[Assembly]), [Total Trips], , DESC, DENSE )`

- Ranks zones based on **trip volume**
 - `ALL()` removes filters to ensure global ranking
-

Filter Condition

Show Top N Trip Zones = `IF( [Trip Zone Rank] <= [Top N Zones Value], 1, 0 )`

- Filters only **Top N zones dynamically**
 - Applied as a **visual-level filter (value = 1)**
-

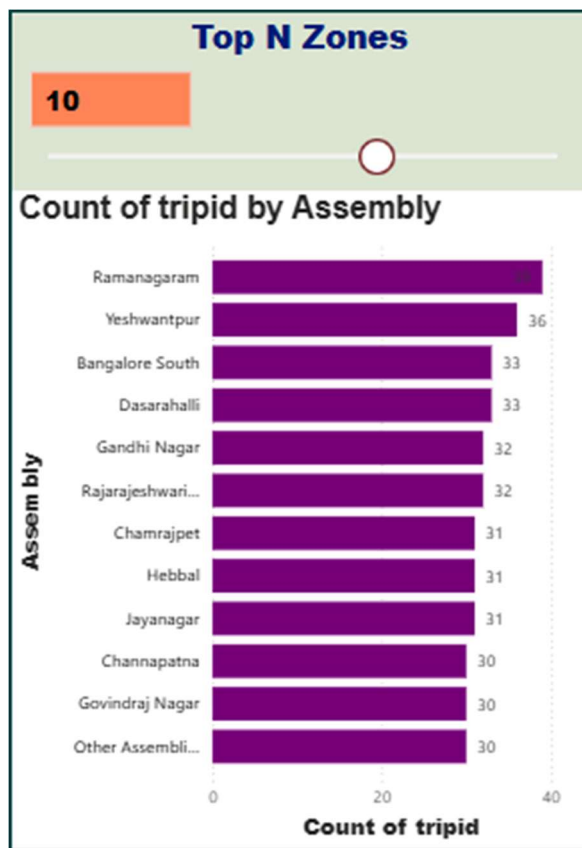
Visualization

- **Bar Chart:**
 - X-axis → Assembly (Pickup Zones)
 - Y-axis → Count of Trips
 - **Slicer:**
 - Controls **Top N Zones dynamically**
-

Why This Filter Was Chosen

- Focuses on **high-performing zones**
- Reduces clutter from low-impact zones
- Enables **interactive exploration:**
 - Top 5 → Quick insights
 - Top 10 → Broader comparison

Observations & Insights



Concentration of Trips

- A small number of zones contribute to a **large share of trips**
Indicates **demand concentration**

Dynamic Pattern Visibility

- When increasing N:
 - Additional zones show **gradual drop in trip volume**
Reveals **long-tail distribution**

Core Operational Zones

- Top zones like:

- **Ramanagaram**
- **Yeshwanthpur**
- **Bangalore South**

Remain consistent across selections

Marginal Contribution of Lower Zones

- Lower-ranked zones contribute:
 - Relatively fewer trips
 - Helps prioritize resource allocation
-

3. Conclusion

[20 Marks]

3.1. Recommendations for Operational Efficiency [10 Marks]

- Based on your findings from the analysis, provide recommendations on how Namma Yatri can optimise its operations.
- This could include strategies for improving resource allocation, reducing cancellations, or optimising ride durations.
- Add supporting dashboards.

Solution:

1. Improve Conversion Rate (Currently 45%)

Evidence from Dashboard

- Conversion Rate = **0.45 (45%)**
- **~55% users drop off** after searching

Recommendation

- Reduce search-to-book friction by:

- Improving **driver availability in peak hours**
- Showing **accurate ETAs & fare estimates**

Impact:

- Even a **10% improvement (45% → 55%)**
→ Can significantly increase completed trips and revenue

2. Reduce High Cancellation Rates (~47–48%)

Evidence

- Customer Cancellation = **48.17%**
- Driver Cancellation = **47.25%**

Almost 1 in 2 rides gets cancelled

Recommendation

- Introduce:
 - Cancellation penalties after threshold
 - Driver incentives for ride completion
 - Improved matching logic

Impact:

- Reducing cancellations by just **10%** can:
 - Increase success rate to ~60%
 - Directly improve conversion and revenue

3. Optimize Driver Allocation by Time

Evidence

- Demand peaks observed at:
 - **0–1 (~50 searches)**
 - **6–8 (~45–50 searches)**
 - **13–15 (~50+ searches)**
 - **22–23 (~48 searches)**

Recommendation

- Increase driver supply during these time slots
- Provide **peak-hour incentives**

Impact:

- Reduced wait time → Higher conversion → Lower cancellations

4. Focus on High-Performing Zones

Evidence

- Top trip zones:
 - Ramanagaram (~39 trips)
 - Yeshwanthpur (~36 trips)
 - Bangalore South (~33 trips)
- Top revenue zones:
 - Bangalore South (~30K)
 - Yeshwanthpur (~29K)

Recommendation

- Prioritize:
 - Driver deployment in these zones
 - Faster matching
 - Surge pricing where applicable

Impact:

- Maximizes returns from **high-value zones**

5. Improve Low-Performing Time Slots

Evidence

- Low demand periods:
 - 8–9 (~28 searches)
 - 21–22 (~30 searches)

Recommendation

- Introduce:
 - Discounts during low-demand hours
 - Driver incentives to maintain availability

Impact:

- Better utilization of idle capacity

6. Address Demand–Revenue Gap

Evidence

- Some high-demand periods:
 - Do not generate proportionally high revenue

Indicates:

- Lower fare per trip or pricing inefficiency

Recommendation

- Optimize:
 - Fare structure
 - Distance-based pricing
 - Micro-surge pricing

Impact:

- Improved revenue without increasing demand

7. Zone-Specific Strategy

Evidence

- Matrix shows:
 - Different zones peak at different times

Recommendation

- Implement:
 - **Zone-time based driver allocation**
 - Localized strategies instead of uniform approach

Impact:

- Better matching efficiency

- Reduced idle time

8. Strengthen Core Funnel (Search → Trip)

Evidence Combined

- Conversion = **45%**
- Cancellations $\approx$ **47–48%**

Strong indication:

- Funnel leakage is high

Recommendation

- Improve:
 - Matching speed
 - Pricing transparency
 - Driver reliability

Impact:

- End-to-end system efficiency improves

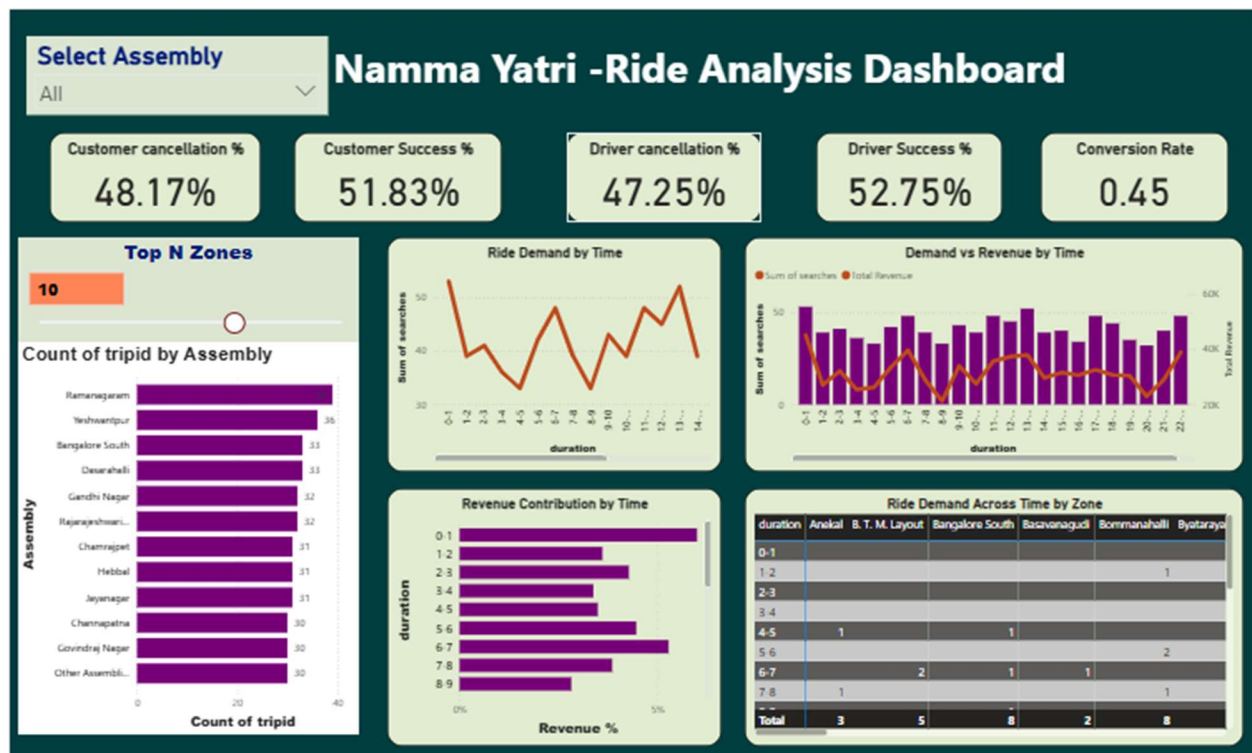
Final Conclusion

- Despite strong demand (up to **50+ searches in peak hours**), the platform struggles with:
 - **Low conversion (45%)**
 - **High cancellations (~48%)**

This results in:

- Lost revenue opportunities
- Inefficient operations

By addressing high cancellation rates (~48%), improving conversion efficiency (currently 45%), and optimizing driver allocation in high-demand zones like Ramanagaram and Yeshwanthpur, Namma Yatri can significantly enhance operational efficiency and revenue generation.



3.2. Marketing and Operational Strategy Improvements [10 Marks]

- Suggest improvements to Namma Yatri's marketing or operational strategies based on your analysis.
- Recommendations could involve promotional efforts, driver incentives, or regional targeting to increase customer satisfaction and service efficiency.
- Add supporting dashboards.

Solution:

1. Conversion-Focused Marketing Campaigns

Evidence from Dashboard

- Conversion Rate = 45%
- ~55% users drop off after searching

Strategy

- Launch "Complete Your Ride" nudges:
 - App notifications when users abandon search

- Limited-time fare discounts (₹10–₹20 off)
- Show **fare transparency + ETA upfront**

Expected Impact:

- Improve conversion from **45% → 55%+**
- Direct increase in completed trips

2. Target High-Demand Zones (Geo-Marketing)

Evidence

- Top zones by trips:
 - **Ramanagaram (~39 trips)**
 - **Yeshwanthpur (~36 trips)**
 - **Bangalore South (~33 trips)**
- High revenue zones:
 - **Bangalore South (~30K)**
 - **Yeshwanthpur (~29K)**

Strategy

- Run **location-based promotions**:
 - First-ride discounts in top zones
 - Office-hour ride passes
- Partner with:
 - IT parks / business hubs

Expected Impact:

- Strengthen dominance in **high-value zones**
- Increase revenue per user

3. Peak Hour Campaigns (Time-Based Targeting)

Evidence

- Peak demand:
 - 0–1, 6–8, 13–15, 22–23 hrs (~45–50+ searches)

Strategy

- Promote:
 - “Zero wait rides during peak hours”
 - Guaranteed driver availability messaging
- Offer:
 - Peak-hour driver incentives

Expected Impact:

- Increase conversion during high-intent periods
- Reduce cancellations due to delays

4. Personalized Marketing (Customer Segmentation)

Insight

- Not all users convert equally

Strategy

- Segment users:
 - Frequent riders → loyalty rewards
 - Drop-off users → targeted discounts
- Use:
 - App-based recommendations

Expected Impact:

- Higher retention and repeat usage

5. Strengthen Brand Trust & Experience

Evidence

- High cancellations indicate trust issues

Strategy

- Improve:
 - Driver ratings visibility
 - Real-time tracking
 - Customer support

Expected Impact:

- Higher customer satisfaction
- Better conversion and retention

By combining data-driven marketing strategies with operational improvements such as targeted promotions, dynamic pricing, and driver incentives, Namma Yatri can significantly enhance customer satisfaction, improve conversion rates, and maximize overall efficiency.

